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Paper 1: The Role of Estradiol in Oxidative Stress in Health and Cystathionine β-Synthase Deficiency

Bibliographic Information

- **Title**: The Role of Estradiol in Oxidative Stress in Health and Cystathionine β-Synthase Deficiency: A Mechanistic Model
- **Authors**: Cruikshank A, Reed MC, et al.
- **Journal**: CPT: Pharmacometrics & Systems Pharmacology, 2025 (in press)
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Quick Take

This paper presents a mechanistic ODE model of hepatic H_2O_2 metabolism that incorporates estradiol and progesterone regulation of key antioxidant enzymes. The model explains why premenopausal females have lower oxidative stress than males/postmenopausal females, resolves contradictory literature on estradiol's effects by showing its influence is nonlinear and context-dependent, and predicts how sex differences in oxidative stress attenuate in cystathionine β -synthase deficiency (CBSD).

Executive Summary

Cruikshank and Reed developed a comprehensive mechanistic mathematical model of hydrogen peroxide (H_2O_2) production and scavenging in the liver, integrating the glutathione (GSH), thioredoxin, and catalase systems with hormonal regulation by estradiol and progesterone. Using this model, they demonstrate that estradiol's stimulation of GPX and GCL creates a nonmonotonic relationship between estradiol concentration and oxidative stress — low estradiol decreases H_2O_2, but higher levels can paradoxically increase it due to GSH feedback inhibition. This nonlinearity, combined with inter-individual parameter variability, explains the wide range of oxidative stress responses reported in clinical studies. The model further reproduces the loss of sex differences in CBSD, where elevated homocysteine inhibits GPX and catalase, overwhelming estradiol's protective effects. The work provides a quantitative framework for understanding sex- and hormone-specific redox biology and its implications for drug development and estradiol replacement therapy.

Scientific Context & Motivation

Oxidative stress is implicated in numerous diseases, and premenopausal females consistently show lower oxidative stress markers than males or postmenopausal females, suggesting a protective role for estradiol. However, clinical and experimental studies report contradictory findings — some show estradiol reduces oxidative stress, others show no effect or even increases. This inconsistency undermines confidence in estradiol-based therapeutic strategies. Additionally, in CBSD (a rare genetic disorder causing hyperhomocysteinemia and oxidative stress), the typical sex difference in oxidative stress disappears. Prior mechanistic models of hepatic metabolism (from the same group) explained sex differences in homocysteine and GSH, but no integrated model existed to explain H_2O_2 dynamics, the combined effect of multiple antioxidant systems, and the paradoxical loss of sex differences in disease states.

⚡ Methodological Snapshot

The authors developed a mass-balance ordinary differential equation (ODE) model of H_2O_2 metabolism in hepatocytes, including cytosolic and mitochondrial compartments. The model encompasses: (1) H_2O_2 production from superoxide (mitochondria) and from peroxisomes/endoplasmic reticulum/NADPH oxidases (cytosol); (2) H_2O_2 scavenging by three systems: glutathione peroxidase (GPX, using GSH), catalase (CAT), and thioredoxin peroxidase; (3) GSH synthesis via gamma-glutamylcysteine ligase (GCL) and glutathione synthetase (GS), with feedback inhibition of GSH on GCL and GS; (4) the methionine cycle and transsulfuration pathway (CBS producing cystathionine → cysteine → GSH); and (5) hormone regulation: estradiol upregulates GPX, GCL, and CBS; progesterone downregulates GPX. Parameters (K_m, V_{max}) were obtained from literature where available; unmeasured V_{max} values were calibrated to match physiological H_2O_2 concentrations ($\sim 0.005\ \mu M$ mitochondrial, $\sim 0.01\ \mu M$ cytosolic) and validated against independent clinical data.

A virtual population was generated by first performing Sobol sensitivity analysis (MATLAB SimBiology) on 48 parameters to identify the top 15 parameters influencing cytosolic H_2O_2. These 15 parameters (including V_{max} for GPX, CAT, GCL, CBS, and transport rates) were sampled using Latin Hypercube sampling within physiologically plausible ranges (Table S1). The population was used to simulate steady-state H_2O_2 responses across estradiol concentrations (0– 1 nM) and dynamic responses across a 28-day menstrual cycle using average estradiol and progesterone curves.

CBSD was modeled by reducing the V_{max} of CBS (cystathionine beta-synthase) to match the ~ 17-fold elevation in homocysteine and $\sim 53\%$ reduction in cysteine seen in patients. Model predictions for CBSD were compared to published clinical data without additional calibration.

📋 Detailed Analysis

The paper's central methodological strength is its systematic use of a virtual population to explore how parameter variability translates into response variability. The Sobol sensitivity analysis (though not detailed in the main text — methods reference the supplement) is a rigorous approach for identifying which parameters most influence H_2O_2 levels. By varying the top 15 parameters, the authors generate a plausible range of individual phenotypes that can be compared qualitatively to the distribution of clinical observations. This approach avoids overfitting to mean data and instead embraces heterogeneity as a feature to be explained.

The nonmonotonic relationship between estradiol and H_2O_2 deserves particular attention. The mechanism is clearly explained: estradiol stimulates both GCL (increasing GSH synthesis) and GPX (increasing GSH consumption). At low estradiol, the synthesis boost dominates, GSH rises, and H_2O_2 falls. But GSH provides feedback inhibition on GCL and GS, so as GSH accumulates, its own synthesis is downregulated. Meanwhile, continued GPX stimulation consumes more GSH. Eventually, consumption exceeds synthesis, GSH falls, and H_2O_2 rises again. This is a beautiful example of how nonlinear biology arises from the interaction of simple regulatory loops.

An important subtlety highlighted by the model is the difference between steady-state and dynamic (menstrual cycle) predictions. During the cycle, when estradiol rises and falls over days, the system does not reach steady state — GSH levels remain relatively stable while H_2O_2 tracks estradiol in a more monotonic fashion. This has practical implications: cross-sectional studies comparing women at different cycle phases may see different estradiol-H_2O_2 relationships than studies examining steady-state hormone replacement effects. The model suggests that the timing of blood sampling relative to the estradiol peak is critical.

The CBSD application is particularly insightful. The model shows that elevated homocysteine inhibits both GPX and CAT, effectively reducing the enzymatic machinery through which estradiol exerts its antioxidant effects. This explains why the sex difference in oxidative stress markers is attenuated in CBSD patients. However, the model also predicts that premenopausal females with CBSD still have lower H_2O_2 than males with CBSD because estradiol's stimulation of GPX (even if partially inhibited) provides residual protection. This subtle prediction could be tested clinically.

One area where the modeling could be strengthened is in the representation of progesterone. Only GPX inhibition is included, but progesterone has many other effects (e.g., on mitochondrial function, on NADPH oxidase) that could influence redox balance. The model's predictions during the luteal phase (when progesterone is high) should be interpreted cautiously given this simplification. Additionally, the assumption of constant H_2O_2 production rates may miss important feedback (e.g., H_2O_2 can stimulate its own production via effects on mitochondrial permeability or NADPH oxidase activation), which could amplify or dampen the predicted dynamics.

From a quantitative pharmacology perspective, this model is well-positioned for integration with PBPK models of estradiol and progesterone to predict tissue-specific exposures and effects. The SBML format facilitates such coupling. Future work should also consider incorporating genetic variability data (e.g., from GWAS on GSH-related enzymes) to create more biologically grounded virtual populations rather than purely computational sampling.

Domain Context

This work sits at the intersection of systems biology of redox metabolism, women's health pharmacology, and mechanistic modeling of rare diseases (CBSD). The model builds on a well-established tradition of mathematical modeling of glutathione metabolism (e.g., Reed et al. 2008, 2015) and extends it to include ROS dynamics and hormonal regulation. In the pharmacometrics field, this type of mechanistic ODE model is less common than PopPK or empirical PKPD models, but it offers advantages for extrapolation beyond observed data — for example, predicting effects of estradiol doses not tested clinically, or of metabolic perturbations in disease.

The focus on sex differences is timely given the NIH mandate to consider sex as a biological variable and the growing recognition that many drugs have different efficacy/safety profiles in men vs. women. The model provides a quantitative framework for understanding one mechanism (redox modulation) through which sex hormones influence pharmacology. This could inform sex-specific dosing strategies, particularly for drugs whose metabolism or toxicity involves oxidative stress (e.g., acetaminophen, doxorubicin, certain antiretrovirals).

For CBSD specifically, the model offers insights that could guide therapeutic development. Current CBSD treatments focus on lowering homocysteine (via vitamin B6, betaine, methionine restriction), but the model suggests that directly addressing oxidative stress (e.g., with N-acetyl-cysteine to boost GSH) might have sex-specific effects that need to be considered in clinical trials. The finding that sex differences re-emerge with treatment (because as homocysteine drops, GPX inhibition is relieved and estradiol stimulation again becomes effective) suggests that treated premenopausal females might show greater antioxidant responses to therapy than males — a potentially important consideration for personalized treatment monitoring.

Key Findings

1. The model predicts that premenopausal females have $\sim 40\%$ lower cytosolic H_2O_2 and $\sim 10\%$ lower mitochondrial H_2O_2 compared to males/postmenopausal females, consistent with clinical data showing lower oxidative stress markers in premenopausal women. 2. Estradiol's effect on H_2O_2 is nonmonotonic: at low estradiol concentrations (≤ 0.25 nM) H_2O_2 decreases, but at higher concentrations H_2O_2 can increase due to GSH feedback inhibition — this explains contradictory literature on estradiol supplementation. 3. In a virtual population spanning physiological parameter ranges, individual H_2O_2 responses to estradiol vary widely, with some individuals showing monotonic decreases, others showing U-shaped responses, reproducing the variability in clinical studies. 4. During the menstrual cycle, mean H_2O_2 decreases as estradiol rises, while GSH remains relatively stable — dynamic behavior differs from steady-state predictions because the quasi-steady-state assumption does not hold. 5. In CBSD, elevated homocysteine inhibits GPX and catalase, reducing estradiol's protective effect and attenuating sex differences in H_2O_2 — aligning with clinical observations that the $NAD/NADH$ ratio sex difference disappears in CBSD patients. 6. The model successfully replicates the ~ 17-fold increase in homocysteine and $\sim 53\%$ decrease in cysteine observed in CBSD patients.

Strengths & Limitations

Strengths

- First mechanistic model integrating all three major H_2O_2 scavenging systems (GSH, thioredoxin, catalase) with sex hormone regulation, providing a comprehensive quantitative framework.
- Grounding in published biochemical kinetics (K_m, V_{max} from literature) rather than purely data-driven fitting enhances mechanistic plausibility.

- Virtual population approach using Sobol sensitivity analysis to identify key parameters and explore inter-individual variability — a sophisticated method for understanding response heterogeneity.
- Model validated against independent clinical data (not used for calibration): fold-changes in CBSD metabolites, menstrual cycle oxidative stress markers, and estradiol supplementation studies.
- Provides a mechanistic explanation for contradictory clinical observations (nonmonotonic estradiol effects, menstrual cycle variability, and loss of sex differences in disease) rather than merely describing them.
- Model code and equations provided in supplementary materials (SBML format), enabling reuse and extension by other researchers.

Limitations (Acknowledged by Authors)

- Model focuses on H_2O_2 as a key ROS and does not include other ROS species (superoxide, hydroxyl radical).
- Constant rates assumed for H_2O_2 production and for glycine/glutamate levels — not dynamically regulated.
- Uses average 28-day menstrual cycle hormonal curves, not individual cycle data.
- Virtual populations not yet calibrated against individual-level clinical data — further validation needed.

Limitations (Expert Review)

- The model is limited to liver hepatocytes; oxidative stress in other tissues (e.g., endothelium, neurons) may involve different regulatory mechanisms and hormonal sensitivities.
- Progesterone regulation is modeled only as inhibition of GPX, which is likely an oversimplification given the broader genomic and non-genomic effects of progesterone on redox biology.
- The virtual population was constructed by varying only 15 parameters from Sobol analysis; true biological variability likely involves many more parameters and their correlations.
- No formal model identifiability analysis or parameter uncertainty quantification is reported — many parameters were fixed to literature values, and the impact of their uncertainty on predictions is not explored.
- Clinical validation relies on aggregate literature data rather than individual patient-level data; model predictions for individuals cannot be directly validated.
- The model does not include dietary influences on GSH precursors (e.g., methionine, cysteine) which are known to modulate GSH levels and oxidative stress.

Generalizability

The model is specific to hepatic H_2O_2 metabolism, which is appropriate for studying CBSD (primarily a liver disorder) and systemic oxidative stress markers originating from the liver. The findings on estradiol's nonmonotonic effects provide a general mechanistic principle likely applicable to other tissues, but quantitative predictions cannot be directly extrapolated. The virtual population approach demonstrates the importance of inter-individual variability, which is a broadly generalizable concept, but the specific parameter distributions are model-specific. The framework and model code are reusable for other questions in women's health and redox biology.

Figures & Tables

- **Figure 1**: Schematic diagram of the mathematical model showing H_2O_2 production and scavenging pathways in the liver cytosol and mitochondria, with hormonal regulation points (estradiol upregulates GPX, GCL, CBS; progesterone inhibits GPX).
  - *Significance*: Provides the structural overview of the mechanistic model, making explicit the connections between methionine cycle, GSH synthesis, H_2O_2 production/scavenging, and hormone regulation.
- **Figure 2**: Steady-state concentrations of key substrates (Hcy, Cys, GSH, H_2O_2) in males/postmenopausal females (Panel A) vs. premenopausal females (Panel B), showing lower H_2O_2 and higher GSH in premenopausal females.
  - *Significance*: Demonstrates the model's core prediction of sex differences in oxidative stress, consistent with clinical observations.
- **Figure 3**: Panel A: Literature data on fold changes in blood oxidative stress markers with increasing estradiol. Panels B-D: Virtual population predictions of H_2O_2, GSH, and GSH/GSSG fold changes across estradiol concentrations, showing nonmonotonic mean responses and wide interindividual variability.
  - *Significance*: Key figure showing the model can reproduce and mechanistically explain the contradictory literature — the nonmonotonic estradiol-H_2O_2 relationship and the broad response distribution across individuals.
- **Figure 4**: Individual virtual subject responses of H_2O_2 (Panel A) and GSH (Panel B) to estradiol at steady state, highlighting different response phenotypes (monotonic decrease vs. U-shaped).
  - *Significance*: Explains the mechanistic basis for individual variability: differences in GSH feedback dynamics drive divergent H_2O_2 responses to estradiol.
- **Figure 5**: Panels A-B: Estradiol and progesterone profiles across the menstrual cycle. Panel C: Literature data on fold changes in oxidative stress markers. Panels D-F: Model predictions for H_2O_2, GSH, and GSH/GSSG during the cycle, showing mean H_2O_2 decreases with estradiol rise but wide variability.
  - *Significance*: Extends steady-state predictions to dynamic hormonal changes, showing different behavior during the menstrual cycle due to non-equilibrium dynamics — important for study design in premenopausal women.
- **Figure 6**: Individual virtual subject H_2O_2 and GSH responses during the menstrual cycle, illustrating how GSH dynamics drive H_2O_2 variability at different cycle phases.
  - *Significance*: Demonstrates that even with identical hormonal curves, interindividual kinetic variability produces widely different oxidative stress patterns during the menstrual cycle.
- **Figure 7**: Panel A: Model-predicted fold changes in Hcy, Cys, and H_2O_2 in CBSD vs. healthy. Panel B: Sex ratios (premenopausal female/male) of metabolites in healthy (blue) vs. CBSD (orange), showing attenuation of sex differences in CBSD.
  - *Significance*: Clinically important: explains the mechanism (Hcy-mediated GPX inhibition overwhelms estradiol stimulation) and implications for sex-specific treatment responses in CBSD.

Code & Reproducibility Assessment

The mathematical model is provided in the Supporting Information (Data S1: equations and parameters with justification; Data S2: SBML file). This allows other researchers to reproduce simulations and extend the model. The virtual population generation uses MATLAB's SimBiology toolbox (sbiosobol function for sensitivity analysis). While the code is not deposited in a public repository like GitHub, the SBML format enables use in multiple modeling platforms (e.g., COPASI, tellurium, Python's libsbml) beyond MATLAB.

Supplementary Materials

Data S1 contains the complete model equations, parameter justification, and virtual population generation steps. Data S2 contains the model in SBML format. Table S1 provides the parameter ranges used for Sobol sensitivity analysis.

Future Directions

1. Validate the virtual population against individual-level clinical data (e.g., longitudinal oxidative stress markers with measured estradiol/progesterone). 2. Incorporate individual menstrual cycle hormonal variability rather than average curves to better capture real-world heterogeneity. 3. Extend the model to include other ROS species and additional tissues (e.g., vascular endothelium for cardiovascular disease implications). 4. Model the effects of oral contraceptives on oxidative stress, as serum H_2O_2 is reported elevated in users. 5. Investigate how genetic polymorphisms in GSH pathway enzymes (e.g., GCL, GPX, GST) modify estradiol's effects. 6. Use the model to predict optimal estradiol dosing regimens for reducing oxidative stress in postmenopausal women. 7. Explore the model's implications for other diseases with sex-disparities and oxidative stress involvement (e.g., nonalcoholic steatohepatitis, cardiovascular disease).

Expert Commentary

This is a well-executed example of how mechanistic modeling can reconcile contradictory clinical observations that have puzzled the field for decades. The nonmonotonic relationship between estradiol and H_2O_2 is a particularly elegant finding — it emerges naturally from the known feedback inhibition of GSH on its own synthesis, combined with estradiol's simultaneous stimulation of both GSH synthesis (via GCL) and GSH utilization (via GPX). This kind of systems-level insight is exactly what models can provide and what intuition alone struggles with. The virtual population approach is appropriate and well-implemented, though I would have liked to see a more formal Bayesian calibration or at least a profile likelihood analysis to understand parameter identifiability. The CBSD application is clinically relevant — understanding why sex differences disappear in disease is important for designing clinical trials that may inadvertently miss sex-specific treatment effects. I particularly appreciate that the authors make the model available in SBML format, which should facilitate adoption by other groups. A natural next step would be to couple this model with a physiologically-based pharmacokinetic (PBPK) framework for estradiol to predict effects of different hormone replacement therapy regimens in specific patient subgroups.

Bottom Line

This mechanistic model provides a rigorous, quantitative explanation for why estradiol's effects on oxidative stress are so variable in the literature — its influence is inherently nonmonotonic and highly dependent on baseline GSH dynamics, individual enzyme kinetics, and hormonal context. For clinicians and drug developers, this means estradiol supplementation strategies cannot assume a uniform antioxidant benefit; patient-specific factors (menopausal status, baseline oxidative status, genetic variation in GSH pathway enzymes) may determine whether estradiol is protective, neutral, or even detrimental. The model also highlights that in diseases like CBSD, where metabolic pathways are disrupted, sex-specific drug effects may be masked at baseline but could re-emerge with treatment, suggesting that clinical trials should stratify by sex and hormonal status.

Figures

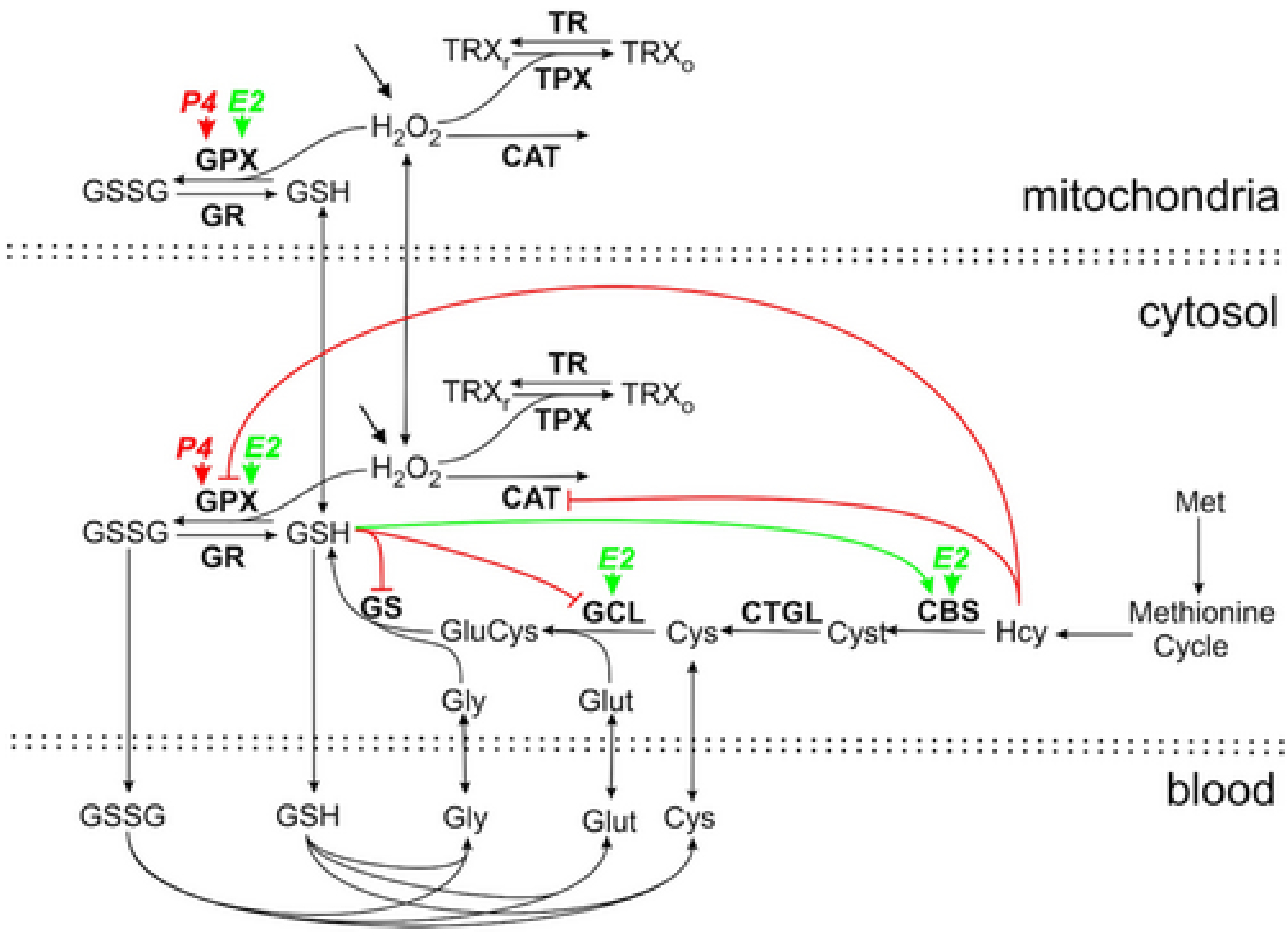

Figure 1: Schematic diagram of mathematical model. Substrates are written in regular text, while enzymes are in bold. Each black arrow represents a biochemical reaction. R

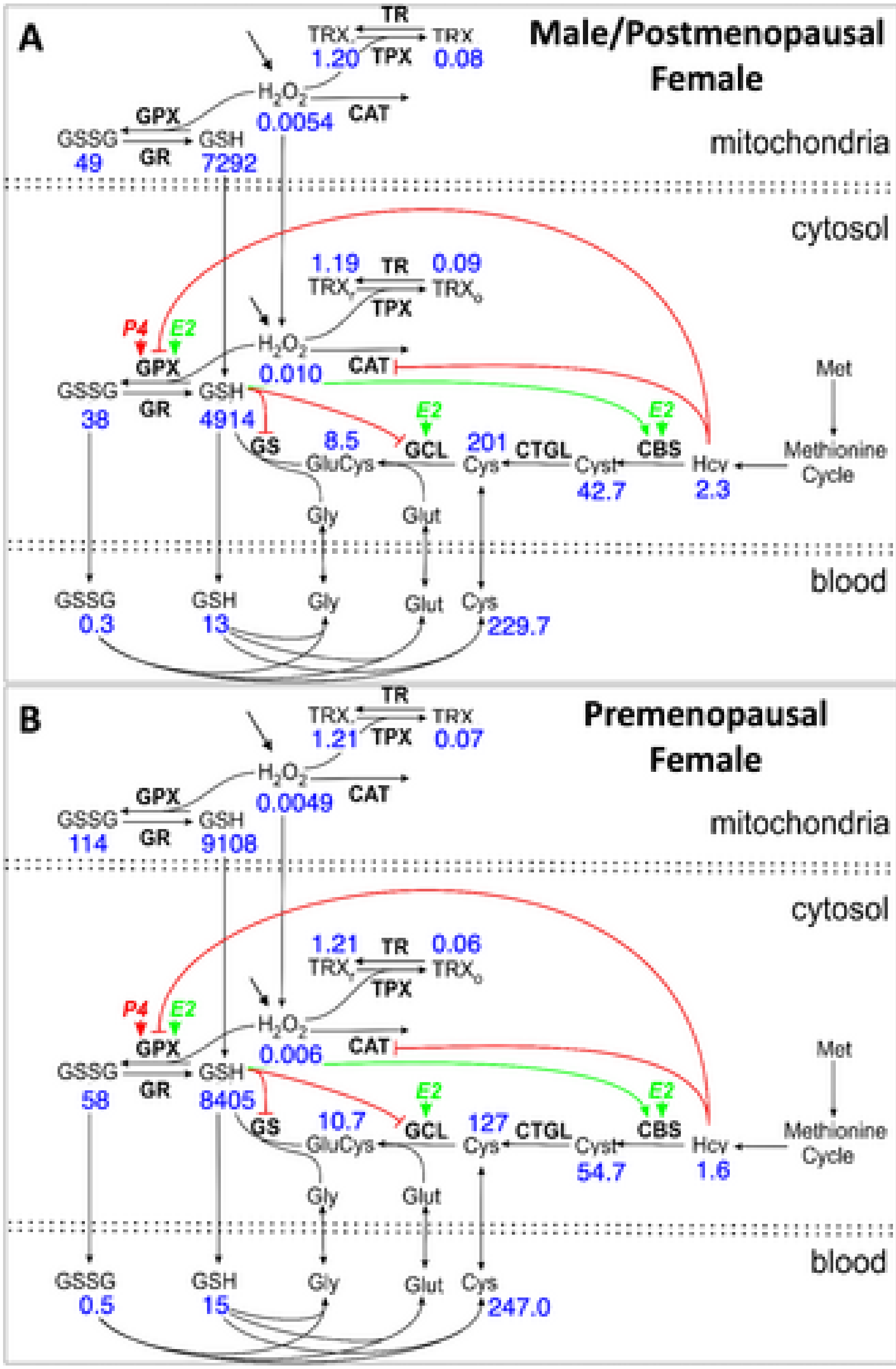

Figure 2: Sex differences in oxidative stress. The blue numbers are the concentrations of the substrates in µM. Panels A and B are the steady states for males/postmenopaus

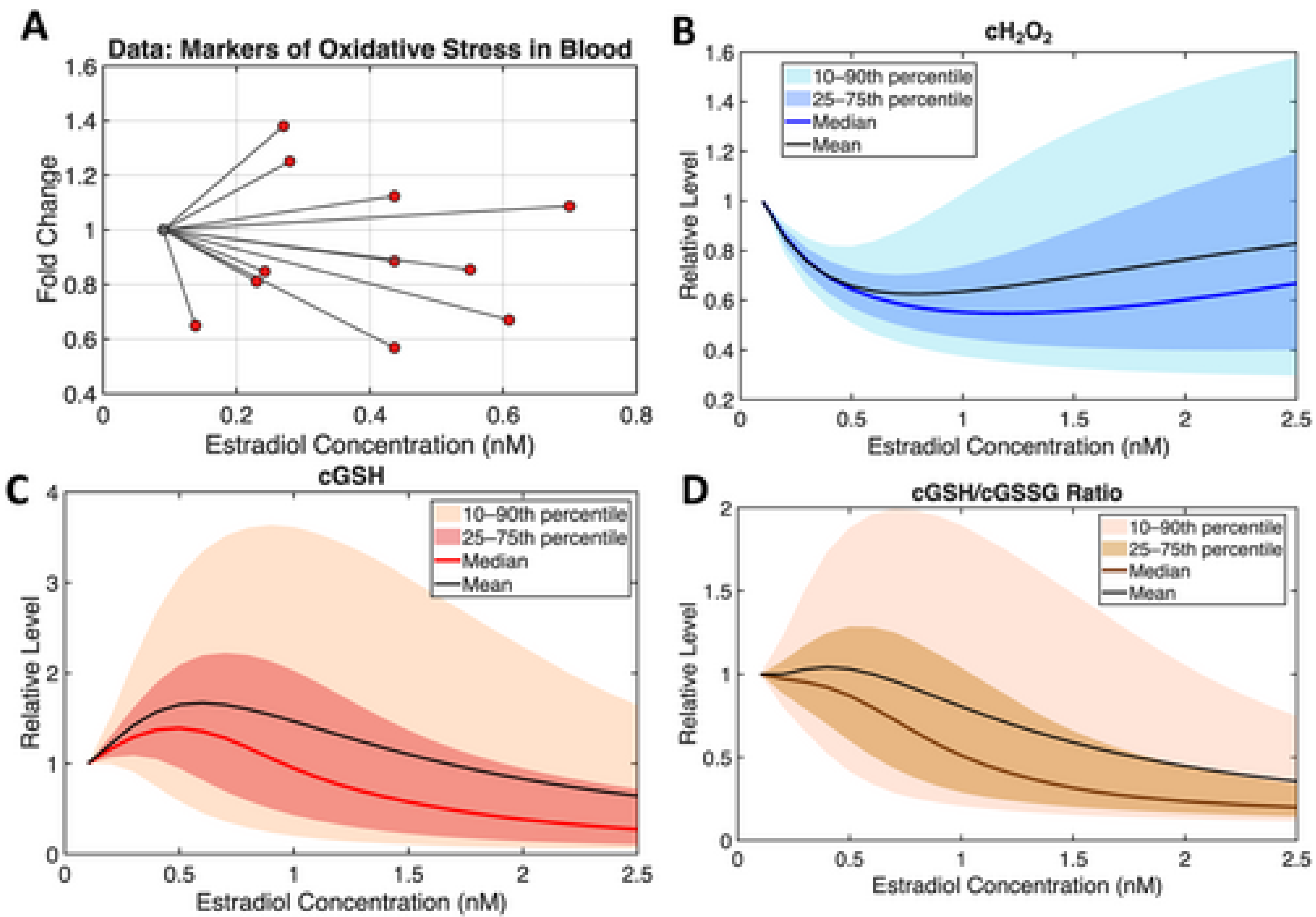

Figure 3: Population model captures variability of Estradiol's impact on oxidative stress. Panel A shows data on the fold changes in oxidative stress markers measured in b

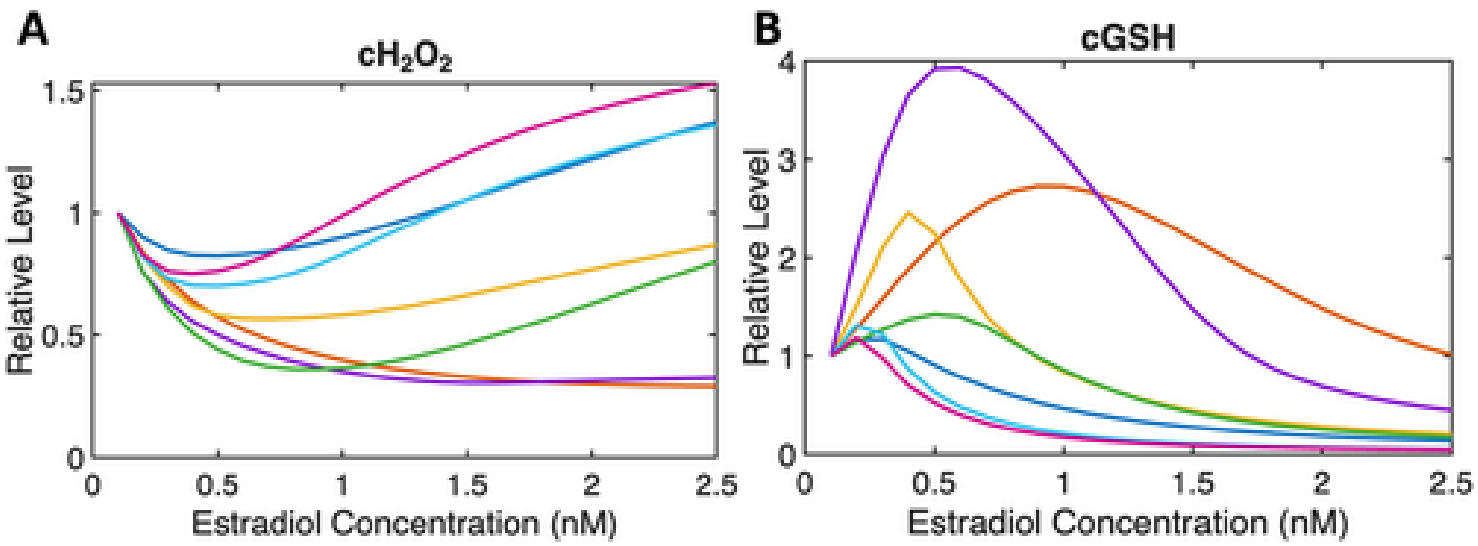

Figure 4: Individual variability in oxidative stress responses to Estradiol at steady state. In Panels A and B, steady-state cytosolic fold changes, relative to an estradi

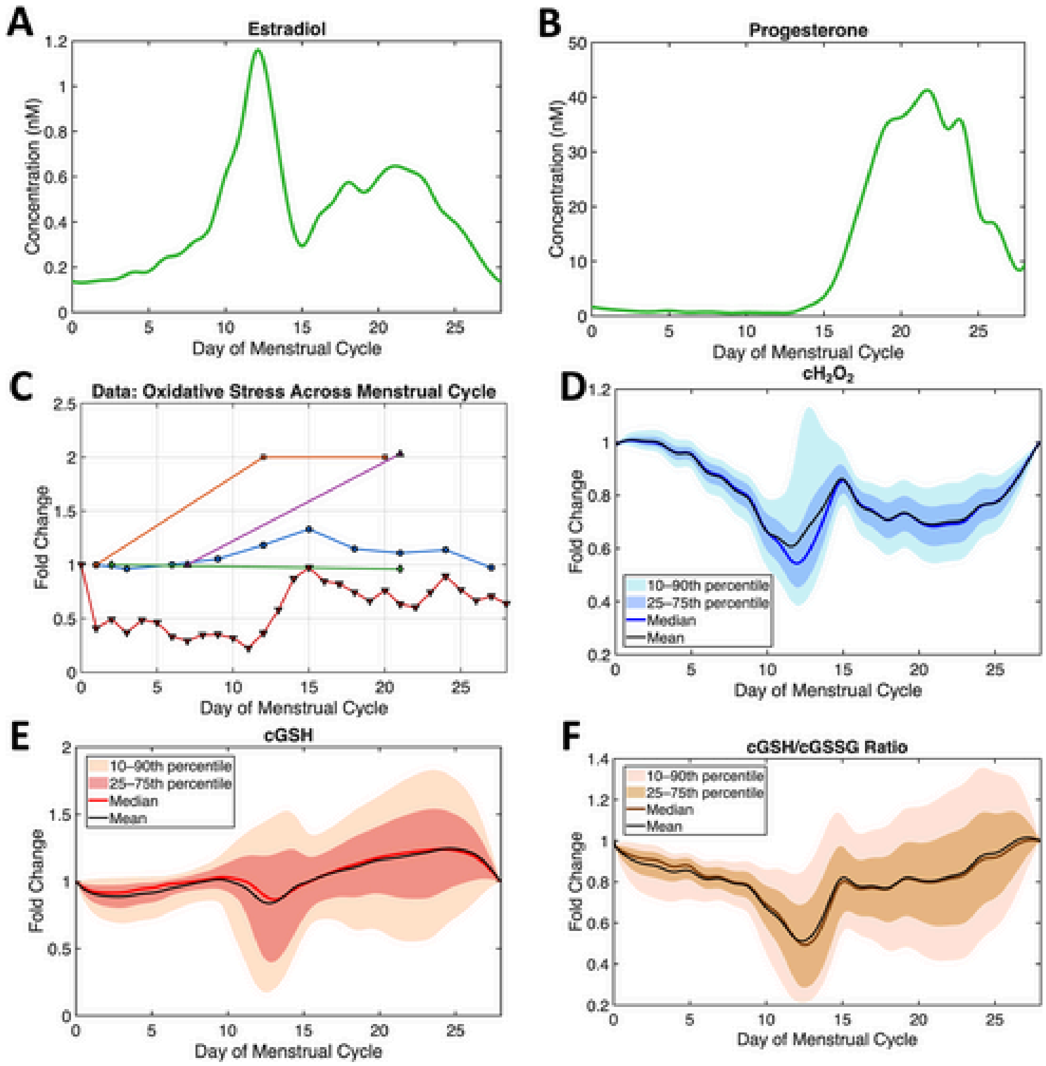

Figure 5: Oxidative stress variability throughout the menstrual cycle. Panels A and B show the mean estradiol and progesterone profiles used in the mathematical model (see

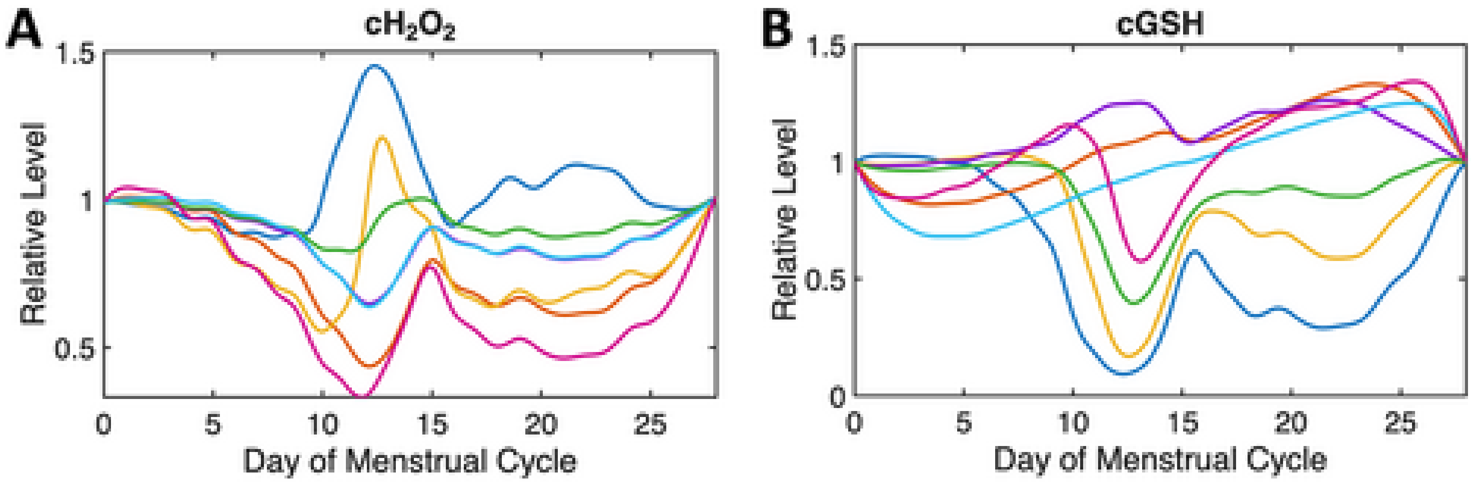

Figure 6: Insights into oxidative stress variability during the menstrual cycle. In Panels A and B, predicted fold changes of cytosolic H2O2(cH2O2) and GSH (cGSH), relativ

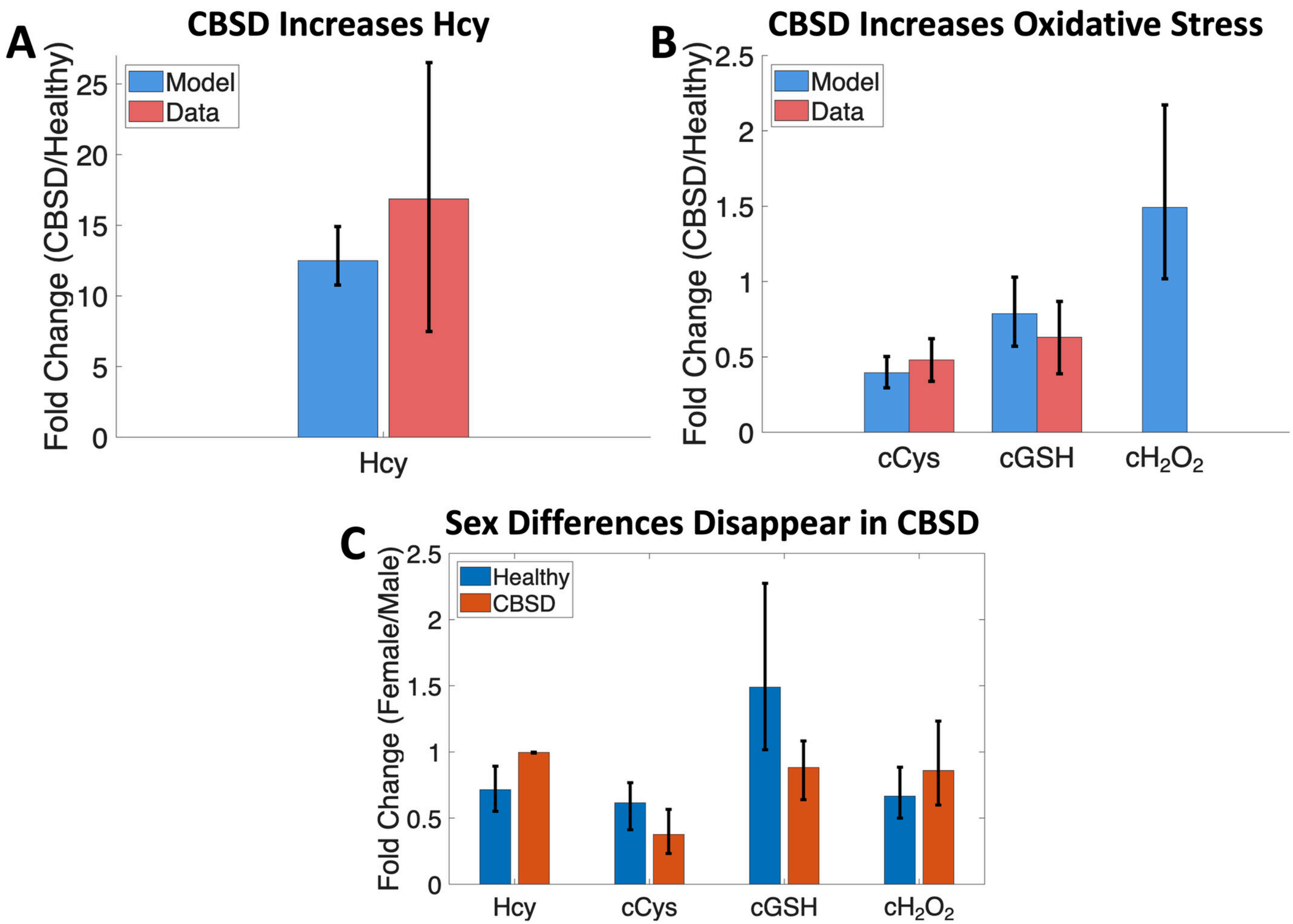

Figure 7: Attenuation of sex differences in cystathionine ß-synthase deficiency. Panel A shows the fold change in homocysteine (Hcy) levels between patients with cystathio